

Salvatore Barbagallo

Data Analyst | Healthcare & Life Sciences Data

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Professional Summary

Bioinformatics and data professional with 7+ years of experience in clinical genomics, specialising in building reproducible NGS pipelines and scalable data workflows.

Designs and implements end-to-end RNA-seq and variant analysis systems using Python, R, Bash, and SQL, with a strong focus on data quality, automation, and workflow reliability in regulated environments. Combines laboratory expertise with computational engineering to deliver robust, decision-ready genomic analyses.

Skills

Core Tools: Python, SQL (BigQuery), R, Bash, Git.

Data & ML: EDA, statistical analysis, feature engineering, classification models.

Data Engineering: ETL/ELT, data validation, pipeline automation, schema design.

BI: Tableau, dashboarding, KPI design.

Languages: Italian (Native), English (Fluent), Portuguese (Fluent), Spanish (Advanced)

Professional Experience

Specialist Biomedical Scientist

University College London Hospitals, Stem Cell Laboratory, September 2021 - December 2025

- Designed and implemented automated tracking solutions (Python/Excel) to replace manual reconciliation processes, reducing inventory errors by 30% and saving 10+ hours per week.
- Managed data tracking and reporting across 11 clinical and ATMP trials, ensuring consistent reporting across parallel workflows.
- Worked with cross-functional teams to improve data quality, reporting accuracy, and process reliability in a highly regulated environment.
- Led the digitisation of SOPs and QA documentation, improving audit readiness and standardising data handling across workflows

Laboratory Scientist

CooperGenomics, London, July 2019 - Sept 2021

- Processed and quality-controlled 500+ NGS samples per week in a high-throughput clinical genomics pipeline, covering library preparation (96-192 samples/run), sequencing workflows, and downstream reporting to ensure data integrity and turnaround targets.
- Delivered 500+ clinical reports per week from NGS data, ensuring accuracy and consistency through structured validation checks
- Developed automated QC validation checks, improving consistency and reliability of sequencing outputs.
- Validated and programmed SPT Labtech Mosquito and Dragonfly liquid-handling robots for scalable library preparation, and trained laboratory staff on automated workflows.

- Produced ISO-compliant sequencing reports with consistent templates and documented checks.

Projects

Google Fiber BI - Market Contact Analysis (*BigQuery, SQL, Tableau*)

- Consolidated **3 regional customer-contact datasets** into a unified BigQuery-ready table of **1,350 records**, covering **85,179 total contact events** across **5 contact types**, and built a Tableau dashboard to identify repeat-contact patterns and service pain points.

Bellabeat Fitbit Data Analysis - Usage & Wellness Insights (*R, SQL, Tableau*)

- Built a Tableau-ready user summary dataset for **35 wearable users** across **15 behavioural and wellness variables**, analysing activity, sleep, heart-rate, and weight trends to support product and marketing recommendations.

Salifort Motors - HR Analytics & Employee Turnover (*Python, scikit-learn, XGBoost*)

- Analysed **14,999 employee records** across **10 departments** and built attrition models using Random Forest and XGBoost to investigate a workforce with **23.8% turnover**, translating model findings into retention-focused recommendations.

TikTok Engagement Analytics - Content Performance (*Python, pandas, Plotly*)

- Analysed **19,382 TikTok posts** to examine how content features related to engagement, identifying highly skewed view patterns and producing data-backed recommendations on content performance.

AWS Three-Tier Solution Architecture

- Designed a high-level AWS migration architecture for two on-prem workloads, a three-tier web application and a Hadoop analytics stack, using managed services including ECS Fargate, Aurora, Redshift, and QuickSight to improve scalability, availability, and operational simplicity.

Certificates

IBM: Machine Learning; AI Engineering; Data Engineering

Google: Data Analytics; Advanced Data Analytics; IT Automation with Python; Project Management; Business Intelligence

Google Cloud: Architecting with Google Kubernetes Engine

SAS: SAS Programming 1: Essentials; SAS Programming 2: Data Manipulation Techniques

Amazon Web Services (AWS): Cloud Practitioner Essentials; Cloud Solutions Architect

Education

Master of Science: Bioinformatics

Atlantic Technological University (Remote), September 2025 – September 2026

Master of Science: Cell and Gene Therapy

University College London, September 2021 – September 2023

Bachelor: Biomedical Science

University of Catania, September 2014 – October 2017